SUVRADIP GHOSH

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SUMMARY

MCA candidate with experience in full-stack development, backend engineering, and machine learning. Developed end-to-end applications through projects involving AI models, API development, databases, and scalable backend systems.

TECHNICAL SKILLS

Programming Languages	Python, Java, JavaScript, C
Frontend	React.js, HTML, CSS, Tailwind CSS, Redux, Zustand
Backend	Node.js, Express.js, FastAPI
Databases	MongoDB Atlas, SQL, SQLite, Redis
AI / ML	PyTorch, TensorFlow, Scikit-Learn, OpenCV, SHAP
Tools	Git, Docker, Clerk, Convex, ImageKit

PROJECTS

InsightATS | AI Hiring & Project Evaluation Platform — Live | GitHub

Tech Stack: React.js • Redux • FastAPI • PyTorch • Redis • BERT • RoBERTa • DistilBERT • SHAP

- Built an AI-powered hiring platform that introduces **Project Complexity Evaluation** alongside contextual Resume–JD matching and resume parsing to rank candidates beyond keyword-based ATS screening.
- Fine-tuned and integrated BERT, RoBERTa, and DistilBERT models, achieving **0.75 F1**, **0.78 Macro F1**, and **0.93 Macro F1** for extraction, resume-jd matching, and project classification tasks.
- Reduced inference latency by **51%** (6.62s → 3.24s) using FastAPI lazy loading, in-memory model orchestration, and Redis caching.
- Reduced API requests by **40%** through shared job-description batch processing while managing asynchronous resume uploads and candidate ranking using Redux.

CodeNest | AI-Powered Cloud IDE — Live | GitHub

Tech Stack: React.js • Zustand • Express.js • Docker • MongoDB Atlas • Redis • Groq Llama-3.3-70B

- Built a cloud IDE supporting **20+ concurrent users**, featuring isolated Docker containerized secure code execution, AI code reviews, flowchart visualization, and **5-second** execution timeouts.
- Engineered an asynchronous execution pipeline using a **25-job queue** and worker processes to improve reliability under concurrent workloads and prevent resource exhaustion.
- Reduced AI response latency to **1–3 seconds** by integrating **Groq Llama-3.3-70B** with Redis-backed caching, in-memory fallback, and secure API rate limiting using SHA-256 cache keys.
- Simplified frontend state management by migrating to Zustand with persistence middleware, eliminating prop drilling and improving maintainability.

Skin Disease Detection using CNN — GitHub

Tech Stack: Python • TensorFlow • EfficientNetB3 • OpenCV • Scikit-Learn

- Developed a deep learning system to recognize skin diseases using the **HAM10000** dataset (10,015 dermatoscopic images), categorizing lesions into benign, melanoma, and non-melanoma classes.
- Improved model reliability through patient-wise data splitting to eliminate data leakage, CLAHE-based image enhancement, hair-removal preprocessing, and transfer learning with **EfficientNetB3**.
- Achieved **90%** test accuracy with F1 scores of **0.89**, **0.86**, and **0.94** using focal loss, transfer learning, and test-time augmentation.

EDUCATION

Master of Computer Applications — 2024 – Present
Sikkim Manipal Institute of Technology — CGPA: **8.25/10**

Bachelor of Computer Applications — 2021 – 2024
Techno India University — CGPA: **8.39/10**

Higher Secondary (Class XII) — 2021
Maynaguri High School — **88.6%**

CERTIFICATIONS

Android Developer Virtual Internship (Google & AICTE EduSkills) — Apr 2024 – Jun 2024
Grade: **Excellent** — Credential

IBM Professional Certificate: Full Stack Software Developer — Credential

IBM Professional Certificate: Data Science and AI — Credential